The Demon's Share

One Bit on the Ledger

Unused Information Is Unheard

Sagawa-Ueda Feedback Demon on a Round Trip

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

One fair bit of information, read off an optically trapped particle at the turn-around of a round trip and fed causally to the return protocol, moves the exponential work ledger <e^(-W)> from 1 toward e^I = 2, capped by the Sagawa-Ueda bound. An equally priced bit that does not predict the work (the sign of x) leaves the ledger untouched at 1.0002: unused or uninformative information is unheard. We measure both branches on one instrument (1.3 million Heun SRK2 trajectories, seed 42): control ledger J = 1.00214 / 0.99731; null sign-bit ledger J = 1.00017; engaged far/near ledger J = 1.0737 -> 1.1270 as the return leverage grows, at 56% of the per-bit ceiling e^I. At the strongest leverage the mean work itself turns negative at DeltaF = 0 (<W> +0.0446 -> -0.0227): work from information. The 0/0 of the ledger -- no bit: J = 1, and a bit held that cannot act: J = 1 -- is removed only when information is spent on the causal arrow.

1. The Instrument: a Round Trip with an Operator

The particle sits in a harmonic trap V = lam x^2/2 with stiffness ramped lambda: 1 -> 2 -> 1 at D = beta = 1 (DeltaF = 0). A round trip with no feedback satisfies the Jarzynski equality exactly (Ch.71):

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<e^(-W)> = e^(-DeltaF) = 1
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At the midpoint the operator reads one bit of x_mid and chooses the return speed: fast, or slow. The protocols are Heun (SRK2) integrators, 0.01 sampling, seed 42; 200,000 runs per branch. No bias: the same bit costs the same coin whether or not it predicts the work of the return.

2. The Fair Bit

Cutting the far/near question at the median of |x_mid| makes the bit perfectly fair by construction:

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median |x_mid| = 0.5682 ; p(far) = 0.4969 = 1/2 ; I = ln 2 = 0.6931 nats
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A fair coin carries exactly one bit of information. Sagawa and Ueda (2008, 2010) convert that coin into a hard ceiling on the feedback ledger:

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J = <e^(-W)> <= e^I = 2.000   (generalized second law)
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Szilard (1929) drew the same coin from a box; Toyabe et al. (2010) measured its face value in experiments: roughly ln 2 k_B T of work per bit -- the price of erasing the coin (Landauer 1961, Ch.68).

3. The Null Meter: Unused Information Is Unheard

The sign of x is measured, priced, and fed to the return speed. The trap is reflection-symmetric, so the sign carries no information about the work:

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sign bit -> return speed:  J = 1.00017, ln J = +0.00017, p(fast) = 0.4999
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Control rows with no bit at all give J = 1.00214 (tau 0.5) and J = 0.99731 (tau 1.0). The null ledger sits exactly on 1.000: a real, priced bit that cannot predict pays for itself and yields nothing. Information is a currency, not a fuel; a coin that buys no leverage is spent in silence.

4. The Engaged Bit: the Ledger Pays

A far particle (|x_mid| above the median) relaxes against a fast return and releases a large negative work; a near one returns almost freely. The demon predicts which, spends the bit on the speed, and the exponential ledger rises with the leverage:

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fast/slow 0.35/2.0:  J = 1.0737   ln J = 0.0711
fast/slow 0.25/4.0:  J = 1.0975   ln J = 0.0930
fast/slow 0.15/6.0:  J = 1.1161   ln J = 0.1098
fast/slow 0.10/8.0:  J = 1.1270   ln J = 0.1196
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J_act/J_control = 1.127; J_act/e^I = 0.564 -- every row respects the Sagawa-Ueda ceiling W of e^I = 2.000 for the fair bit. Crucially, the mean work itself turns negative at the strongest leverage even though DeltaF = 0 (<W>: +0.0446 -> +0.0156 -> -0.0092 -> -0.0227): the demon harvests by predicting, and the harvest is capped per bit by the coin.

5. Books Close with the Coin

No bit: J = 1. A bit in hand that cannot act: J = 1. A bit that predicts the work: J rises toward e^I and stops there. The removal of the 0/0 -- the ledger is 1 both with no information and with dead information -- is the entire content of the generalized second law <e^(-W-I)> <= 1 (Sagawa-Ueda 2008/2010; Parrondo-Horowitz-Sagawa 2015). The demon is not a free-energy source; the demon is a portfolio manager. The information is the payment, the ledger is the receipt, and the reversible corner of Ch.73 is priced per bit at the interest rate Ch.71 measured.

Main Results

THEOREM

Causal Information Raises the Ledger, Bounded by the Coin: Let J = <e^(-W)> be the exponential work ledger of a round trip (DeltaF = 0, so J = 1 with no feedback). A bit of measured position fed causally to the return protocol gives J <= e^I with I the Shannon information of the bit; a bit uncorrelated with the work leaves J = 1. Measured: null J = 1.00017; engaged J up to 1.1270 = 0.564 e^I.

COROLLARY

The 0/0 Resolution: no bit and a dead bit are both J = 1; the removable value is realized only by information spent on the causal arrow.

COROLLARY

Work from Information: at the strongest leverage <W> = -0.0227 at DeltaF = 0; the demon extracts work by prediction, capped at ln 2 k_B T per bit (Sagawa-Ueda; Toyabe et al. 2010).

Conclusion

The exponential ledger is the universal balance sheet of Chapter 71; here a demon signs one fair bit against it. A bit that does not predict -- one bit more expensive, identical ledger -- demonstrates the null branch: unused information is unheard. A bit that predicts the work moves the ledger 1 -> e^I and stops, never above, at 56% of the ceiling measured. The demon's share is not free energy: it is the repayment of a loan the noise already granted, signed by the particle, priced per bit, and audited by the coin.

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